ActInf GuestStream 100.1 ~ Yu Kanazawa: The free energy principle, langu

GuestStream | Yu Kanazawa: The free energy principle, language learning, and education | 1:29:40

Hello, welcome everyone.

It's March 14th or 15th, 2025, and we're in Active Inference Guest Stream 100.

So congrats on the triple digits.

We're here with you, Kanazawa, and we'll be discussing the free energy principle, language learning, and education.

So we are looking forward to a presentation followed by a Q&A.

If you're watching live, please feel free to write any questions in the chat.

And thank you, Yu, for coming on and looking forward to your presentation.

Thanks again.

All right.

Thank you so much for inviting me to this opportunity.

And also, I would say it is not without some trepidation that I'm here because basically

I'm an applied linguist or foreign language education researcher.

And also I had a look at those YouTube page.

And usually there are some like mathematical views or some scientific views.

But what I usually do is, of course, some more pedagogy oriented or some psychology.

But I do my best.

And also, yeah, this is pretty much open interdisciplinary framework.

So I try to use it or incorporate it or getting some insight or like a mixing insight from applied linguistics or second language acquisition, especially the instructional kind.

And this new, very influential and promising idea of active influence of free energy principle.

So I get to it.

Okay.

Yeah, maybe I need to acknowledge this first.

So this work or my study project supported by some Japanese governmental Akenhi project.

So I need to acknowledge that first.

And yeah, so as I said, I had a look at those free energy principle literatures or some YouTube sites.

But usually people don't talk about education or like language education, especially.

But here's my background.

So basically, if I was asked what I'm doing, what do I do?

Well, the research field, of course, is more about some second language education or applied linguistics.

That's my field.

But there are a lot of different approaches there.

This field is pretty interdisciplinary oriented.

And one thing I have done for long is number two is vocabulary or formulaic sequences.

That's more like learning foreign languages.

The learning vocabulary, like what's that different aspects of vocabulary and stuff.

And I published a book or like book chapter, mostly in Japanese though.

Another topic that I started when I was a doctoral student is emotion, actually.

Emotion is another pretty big topic, field in that I'm a research topic, especially from psychological, philosophical approaches, and also including cognitive science.

Actually, the first opportunity that I came to know this free energy principle is from emotion literature, actually.

So that's another thing.

And also the third thing is pedagogy, foreign language education, especially something called active learning or dialogic learning,

or some more like learning method, the student-centered learning method that is more engaging, or that's triggering student motivation.

So these three things basically I have been doing.

It's kind of expanding more, but basically this is it.

And next one.

Yeah, also another thing here.

So today's presentation, this is based on this paper,

The Free Energy Principle and its Implication to Language Learning and Education,

a theoretical paper that I published last year, 2024.

So it's, I talked about 4E cognition, prediction, accuracy, complexity, trade-off,

interesting motivation by epistemic emotions, 4Skills, and other things there as well.

So basically it will be there.

And also this QLs or this ULL there is basically where you can learn more about that in case if you are interested more.

And yeah, I'm from Japan, based in Japan, Osaka, the Kansai, the Western area,

which is, you know, the second biggest part compared to like Tokyo, the Eastern part though.

So in Japan, there has been a lot of focus or publication study about Free Energy Principle as well.

Maybe it's nice to briefly mention these things there too.

So especially like Professor Inui, he's based in Osaka, North Osaka, same place as mine as well.

Also applying it, some introduction to Free Energy Principle published recently.

Also explaining, applying it to emotion study.

That's the second book on the left side.

And also there's some translation there as well.

Yosoku Suru Koko.

This is the How We Predictive Mind book.

That's also translated into Japanese.

And also the right side, the green one, the active inference.

This is like Parr et al. 2022, which is an awesome book.

And I think it's used as a textbook for this, you know, the reading group as well.

So this actually, after reading this book or these books, especially this Parr et al. 2022,

I had the idea that maybe I should use some insight somehow to language education.

So this has been some direct motivator.

That's leading to my some ways of thinking and also ideas out here.

And also it's not directly about language education still, but there has been a lot of Japanese researchers investigating further into this free energy principle there as well.

So for example, on the left side up, making some, oops, maybe it's actually just look this way.

Yeah.

So forward a model of design process based on free energy principle.

So combining it with Charles Sanders Peirce's theory, which is another favorite theory of mine, actually, though.

And also emotion, psychology or science.

University of Tokyo Energi Sawa Sensei's studies out there about free energy, applying it into emotion, valence emotion or epistemic emotion.

That's another topic that I talk about later.

And also even in those second language education on the right side up, some neuroscientific second language empirical study conducted.

And briefly, the idea about free energy principle or predictive processing is also mentioned and also integrated into analysis.

And something very recent, just a few days ago, the right side down, like system 0, 1, 2, 3, quad process theory of multi-stay.

Time scale embodied collective collective cognitive system.

This is very pioneering, interesting idea about some Kahneman's system 1, 2, expanding it using some Henri Berksonian idea about time.

And also free energy principle based predictive processing, which is a preprint, but authored and studied by leading researchers in Japan.

Anyway, so briefly I talked about second language.

So maybe I should talk more into my own theory or ideas, conceptual analysis.

So I'll just put in the FLE.

So let me just put in the FLE.

This is an interdisciplinary field where various theories and insights have been applied from various varied academic domains, such as cognitive science, but also from psychology, philosophy and even beyond.

And even beyond.

So it's like some intersection of different academic field.

So it's kind of interesting, exciting field actually.

And one noteworthy theoretical framework in the state of the art cognitive science is, of course, the free energy principle or active inference.

In France recently, it's also known as it was proposed by Professor Carl J. Princeton.

And I think if I'm correct, it's mean in mid 2000.

So in this century and has been elaborately and cross-disciplinary developed ever since.

Like very productive.

So each week or each day, there are new free energy related publications somewhere.

And making it a promising, groundbreaking, unifying theory of the brain, or even further, the dynamics of living systems.

In this lecture, the presenter, me, Yukon Ozawa, introduces some key concepts of this free energy principle FAP framework, based mostly on this part at all 2022.

Understanding from it in a greatly simplified manner.

So this is one of the, I think the rare attempt to explain it without using mathematical formula.

So, so I don't talk into, I don't go into mathematics or those mathematical aspects of the theory, but some what implication it can, it can be derived from the theory and applying it to language education or like instructive signal acquisition.

Or like teaching.

And par at all.

So this is it.

And another great thing is, it's open access.

You can download the PDF from MIT press website.

And also it's kind of off topic, but in my field applied linguistics, there's more and more.

I would say some people's interest toward open science or the need of equality, kind of social justice for information.

In that sense, open access is highly valued.

Very good introduction.

is all free for economics is to make this bookkeeperkeeper

off topic but in my field applied linguistics there's more and more uh i i would say some

people's interest toward open science or the need of you know so that equality uh kind of social

justice for information in that sense open access is highly valued and so this book is in a sense

also leading it in that aspect has a very good introduction and it's all free for anyone in the

world if they have internet connection and uh this uh is from the paper action perception loop so you

have like the mark of blanket some uh uh usually for non-scientific people is like exotic word there

but you know the internal system external systems are divided uh like separated uh using this uh
blanket and then uh the asian side and the world side but there are some interaction
there internal state through like sensory states or like perceptions uh of course there are some
input based uh interaction there and also through action uh it's also still possible to go influence
from internal state to external state a little bit of another kind of off topic there too but uh this
book like i think naive 2025 another recent mit press uh book about applying uh free energy principle to
like like life's or like a more like broader uh perspective of like biology i think uh this idea
about mark of blanket realism has been actually uh questioned or maybe some better ideas proposed actually
which the book i i was reading yesterday and maybe i was reading too much so maybe i should have prepared
more for slides at the expense of my sleep a little bit but yeah well uh there is of course these
theories are uh change slightly changing or modified and stuff so maybe uh what i written here is based
on what i understand last year at last year so uh diving into the first aspect uh for e-cognition
um foreign language learning education and free energy principle well uh ai picture usually you know the
generous and nice image ais are usually useful so sometimes i do that um yeah so uh for e-cognitions
there so that when i first read uh free energy principle i thought oh this is pretty much harmonious with
or resonate those four cognition for e-cognition idea uh although it was not explicitly written down there
as well but you know it's famously i think for uh uh cognitive psychologists or scientists it's kind of obvious
but uh there are four e's out there embedded cognition embodied cognition inactive cognition and extended cognition
i don't talk too much into these things but actually i'll just point out that these are related both to
free energy principle and also to foreign language education as well and for those free energy principle
and for e-cognition like inactive and others those like naive that 2025 is a perfect another recent example
of the book but you know just these things as are written they're implicit generalized so basically it's like
embedded so we are in context of the environment so we're not just a single uh being in vacuo uh just
separated agency you know but we are uh the environment context having great impact on the
cognition and also that's also in foreign language education uh literature too that was also pretty much
recognized actually interaction learning context in the past it was more like input reading listening but
now like interaction is more important or like what to do how to ensure uh interaction that all the
factor of the environment these are also highly studied embodiedness the body or using the body
traditional example in language education like tp earl total physical response but also recently
there's some shadowing practice like actually you pronounce something uh there's some skill acquisition some
drill but that can also be more social or like in interactional and also inactive approach there as
well some task-based language teaching project-based these new pedagogical approaches are pretty much
related to this inactive idea those like active agency or productiveness of the learners the basis of these
ideas and even extended cognition there which is of course recently those ict tools there

uh or chat gpt or generative ais or even using uh some of those tools they will lead to how to instead
of just whether to use it but like how wisely to incorporate it into language learning that's also
highly highly studied topic like almost all conferences there are those like how to use ai's wisely
topics in language education there yeah several like books so like fully cognition like uh 2020 2018 like
new one at all as another great introduction plus more explanation about four e's and naive 2025 that's
another great book just talking about uh free energy and for recognition and some examples of language
education like some shadowing book kadota 2019 is another one like introducing some more some i would say
embodied aspect of uh pronunciation or like listening so for a skill uh fluency fluency training and
forthcoming next year some generative ai how to use it well for language education from like multi-literacies
also these books and uh papers are published kind of every day i would say
so well just talked about some relevance of four e's free energies and also education so let's move
on to some omnipresent prediction the idea of free energy itself well um example of reading in foreign
language education well uh usually those foreign language education literatures or like second language
acquisition approaches there are four skills people often talk about four skills are listening reading
speaking and writing basically and of course like reading is one of the most uh fundamental aspect
especially for english as a foreign like foreign language learners if they need to use it in their daily
life more speaking or listening is important but especially if you're learning as a foreign language
because reading is one of the most important skills a dominant view of reading in second language is that
reading essentially concerns input-based data-driven decoding and that additional attention of
consciously making predictions and inferences will turn its passive nature toward more active and
top-down processing there's been several models like bottom-up reading top-down reading but
top-down reading models while they are like guessing game model all those things but recently like
a lot of theorists or practitioners or scholars agree that reading is basically based on the bottom-up
process the text first and then you decode it and that gets your brain basically and then at that stage
there is also some prediction or thinking those top-down processing also comes into play but basically
it's input like listening and reading are input and bottom-up processing is the basic mode of reading those
top-down concept driven perspectives can be added but the basic nature is bottom-up
but is this true well kind of drawing some insight from free energy principles can we really say that
those input is really perception input or like some input is first and then action understanding later
maybe it doesn't have to be so latest cognitive science suggests otherwise the fundamentality of
predictions prediction is kind of like top down processing your brain or cognition first and then
that's kind of influencing what you perceive and that is even unconscious those idea about unconscious
prediction that's pretty new in second language education i would say or acquisition theory and just
you know there's several examples the first one uh you read this sentence right kind of stuff most
readers will be able to understand what's written in the sentence with jumbled letters
uh that's because and also even sometimes you may not realize that it's jumbled letters there are
lots of similar examples in different languages there as well but so in other words like we don't really

look into those letter by letter and construct uh like letter to word word to sentence in a bottom-up
manner we rely greatly on those kind of system one based prediction kind of intuitive process i would say
another example is masked semantic priming mask priming is some psycholinguistic approach that you that
priming is like you look at something first you look at something next and the first thing you look at
somehow influences the second thing uh like for example uh you see the color first and then you see some
name of fruit for example like red color and for example apple i would say maybe that if that's
congruent maybe the processing is faster like faster reaction time shorter reaction time or something
but usually so those things may happen with masked condition mask is like the first prime is presented only
like uh 50 milliseconds or something so you are not aware of it you are not consciously aware that
something was presented but still it was even at the meaning level something that you unconsciously processed
first may have an impact on something that you process next the performance the reaction time and stuff so
unconsciously we are paying attention to many things it's just that we are not aware of it so that may need
some paradigm shift or with different ways of thinking about just input processing maybe it's not just
input behind input there are a lot of active inference or active perception there and those helm helm
holtzian idea of perception as unconscious inference out there just i'm citing that book though perception is
not a passive outside in process in which information is extracted from impressions on our sensory
epithelia from out there it is a constructive inside out process in which sensations are used to confirm or
this confirm hypothesis about how they were generated so future like foreign language education theories and
practices may well recognize the active nature of perception i try to operationalize it
so i'm going to see some books and uh hobby book of course also in japanese translation as well also
kind of related which i use as the textbook uh in several of my university classes uh barrette the seven
and a half lessons about the brain also the importance of prediction uh and our brain or cognition is
pretty well documented or explained in this popular science book
all right so now going further into free energy itself free energy minimization
so what's it about so free energy minimization so what this is about the energy principle is not the
only model that theorizes the predictive nature of perception so there are a lot of predictive processing
literature and like uh also attempts out there that's related but not necessarily free energy
what makes it outstanding is its extension from perceptional proper perceptional inference to active
inference and resulting unifying theory of perception and action under the same underlying mechanism of free
energy minimization so it's just kind of extending kind of bergsonian way of saying it but extending
perception into action and these actually under the same mechanism so off topic but this idea is pretty
much echoing the henry bergson's idea about perception and memory which is also now highly studied
the free energy principle posits that minimizing surprise or surprisal
is the furnace is the furnace is the furnace is the furnace is the furnace is the furnace is the

avoiding global thermodynamic disequilibrium, which is like death or something.

So minimizing surprise, minimizing unexpectedness.

And this is done by minimizing the proxy, the free energy.

And here, especially, it's a variation of free energy.

Free energy, which is an upper bound on surprise value given a generative model,

is a function of the organism's internal dynamics and its sensory and active states.

It's not just either, but like both.

An example for more kind of intuitive example here.

Two ways of minimizing free energy.

Perception.

One is perception changing.

Perception or changing yourself.

The other, action or changing the world.

And in my paper, I wrote about an example of a traveler going to a foreign country where people speak some completely unknowingly.

unknown new language.

You hear suddenly someone speaking in your mother tongue in a remote foreign country that you expect nobody will speak your language.

What should you do?

Well, one possible thing to do is update your belief based on your perception.

In other words, you can maybe think about, oh, some people here speak your language.

Maybe there are other travelers from your country.

Another way to go forward is taking action.

For example, walking closer, finding who was talking, who was speaking.

And then you can even talk to the person to make sure that, oh, surely there is someone who speaks your language.

Or it could also be some coincidence.

Maybe just one word you kind of misheard as your mother tongue phrase or something.

Actually, both are out of the same principle of minimizing surprise there.

Both approaches can be taken.

Not just either, but kind of both.

Free energy can also be formulated and defined as the difference between complexity and accuracy.

Where the goal of inference is minimizing complexity while maximizing accuracy there.

So here, there is a trade-off between accuracy and complexity.
And as Einstein noted, as simple as possible, but not too simple.
So you should try to make it as simple as possible to make it tangible or manageable.
But if you do that too much, it's no longer useful.
And they are in trade-off relation with each other, as this formation suggests.
So what can foreign language education or learning can learn from these facts?
Well, at least two things.
One, similarly to the relation between perception and action in the free energy principle,
those input processing, reading, listening, and output processing or production, writing, speaking,
may also be conceptualized as inseparable and explained by the same underlying mechanism.
In the past, people used to usually just divide these four things.
I remember when I was undergrad or something, someone just asked me, like some graduate student
talked to me, like, which are you interested in?
Listening? Reading? Speaking? Writing?
Speaking? Actually, my idea was like, oh, I shouldn't just pick one.
And so in that sense, of course, maybe, of course, there are different theories for each,
but just behind it, there must be something which is common, some underlying mechanism,
or even interaction that is like a mutually influencing with each other in very subtle manner.
So we should pay attention to those subtlety or, I would say, let me look for a better word,
those mutual permeation, I would say.
Maybe permeation is a good word to go for.
Interpenetration, permeation of different things, seemingly different categories.
And also, a trade-off of accuracy and complexity is pretty much echoing this idea of CAF, or CAF,
which is highly studied in second language acquisition.
Complexity, accuracy, and fluency.
Well, these three kind of criteria often used to measure, like, second language production,
or, like, estimation or assessment.
And, of course, like, higher these values, OR, which is good,
but how they are related, actually.
And also, sometimes, their trade-off relation, for example,
focusing too much on fluency, maybe at the cost of accuracy or complexity.
All these things can happen for others there, too.
And how to kind of scientifically kind of fundamentalize these ideas.
Free energy principle literatures have a lot of scientific insights
about the nature of complexity, the nature of accuracy,
and even, recently, the nature of fluency, as well.
So, maybe these insights could be somehow incorporated,
or used for theory for these CAF aspects in second language acquisition, as well.

So, that was mostly about those, the first part of free energy,
but, which is, like, a variation of free energy.
But, it's not just that.
Something deeper.
Into the temporal depth.
Well, minimizing surprise of the present
could end up in stubbornly persisting to its own equinish.
So, just minimizing surprise now,
maybe it would just end up no exploration.
Maybe you're just trying to keep your homeostasis only.
But, like, life, or, like, a living organism, which is different.
Like, there is some evolution, or development, learning.
Also, pretty much well discussed in the recent NAV 2025, as well.
Such a strategy may,
such a strategy like just persisting to your homeostasis,
may be maladaptive in the long run,
because it essentially disfavors newness.
Trying to keep itself within the comfort zone,
unable to cope with dynamically and unpredictably
changing situations in the future.
So, just trying to minimize surprise, surprise all now,
maybe you will not adapt to the new situation in the future.
And, this, for instance, et al.,
and also active inference researchers,
they incorporated into the model.
Generative models endowed with temporal depth,
different time scale, like future-oriented thinking,
and hierarchical temporal structures are called for.
And that resulted in extending the scope
of the original variation of free energy
to the expected free energy, EFE.
And again, EFE is also well explained in Power et al. 2022,
so not talking too deep into it,
but two ways of minimizing those expected,
kind of future-oriented free energy,
as well, there as well.
One is kind of exploitation idea,
which is somehow similar to the variational one,

like maximizing utility
based on the prior preferences,
or pragmatic value,
something you can use,
you keep on using,
or kind of exploiting on it,
but it could result in kind of using up the resource though.
But at the same time,
not just that,
you could also do it this new way,
exploration.
You maximize information gain
by reducing uncertainty
about the cause of valuable outcomes,
which has an epistemic value.
And when some environmental situation change,
this exploration may end up pretty important
because you're not just sticking
to your old way of doing it.
And these two,
exploitation versus exploration,
that's automatic.
The balancing is automatic
for exploitation and exploration.
It's mediated and motivated by,
well, this is my way of phrasing it,
but epistemic emotions there,
of course,
surprise,
free energy,
and like boredom,
result of exploitation,
maybe you may feel boredom,
which will result in
some internal drive
to look for newness,
and then the curiosity toward exploration.
And boredom,

curiosity,
these concepts are pretty well
studied recently,
new handbooks,
more and more papers,
scales,
and these can be categorized
into this category
of epistemic emotions.
And epistemic emotions
and temporal depth.
Well,
monitoring free energy dynamics
elicits emotional states
at multiple levels.
And these emotional states
may play crucial roles
in free energy minimization.
Like basic forms
of hedonic emotions,
such as happiness
and fear,
these may help
biological agents
to adapt to changes
in the world
via variational
free energy minimization,
which is also pretty much
true for any organism,
like at any
evolutional level.
But in turn,
epistemic emotions,
which are related
to kind of more
higher cognitive processes
of more higher,

more complex organisms,
even some consciousness
involved probably,
conscious knowledge exploration,
exploration,
and acquisition.
This may provide
a tangible interface
between deliberate
intellectual learning
and expected free energy
minimization
in rich,
temporal depth
in a more
future-oriented manner.
So,
what implications
are there
for foreign language
education and learning?
One thing,
well,
since language learning
is deliberate
and effortful
higher cognitive process,
of course,
and,
you know,
it's usually
it may start
when you are older
or second language
processing.
So,
you need to
exercise more of your

some intention
on deliberate
cognition to learn.
And here,
the temporal depth
should be considered
both in pedagogical
planning and research.
Like,
for example,
complex dynamic systems theory,
that's pretty big
in applied linguistics now,
or approaches
of quantifying depth
in foreign language
education.
For example,
like deep active learning.
So,
what constitutes
deep?
So,
like,
providing some
theoretical basis
behind
what does
depth mean?
Also,
intrinsic motivation
is explainable
by expected
free energy
minimization
by epistemic
value maximization,
which is situating

foreign language
motivation theories
in more scientific
contexts.
Motivation
is another
pretty big
applied linguistics
theory
that many
researchers
and educators
are interested
in.
There are a lot
of models,
but there are
some
validational
crisis
of several
models
and stuff.
But,
maybe,
based on
this,
what is
interesting
motivation?
Well,
this is actually
coming from
those epistemic
value maximization
from the
minimization
of expected

free energy.
This may have
been some
theoretical,
some scientific
background
of what
is interesting
motivation,
which is not
just only
from philosophical
or observational
findings.
Well,
just kind of
like off-topic,
but,
like,
here are some
examples,
like deep
active learning.
Active learning
is like
learning,
not just
teachers
lecturing,
but students
doing discussion,
group work,
but,
recently,
more,
just active
is not good
enough.

Maybe outside
behaviorally active,
but internally
learning is not
deep,
that's the problem.
So,
there's deep
active learning
is proposed,
that's leading
to deep learning,
understanding,
and engagement,
and also another
pretty big field
in my second
language acquisition.
Psychology language
learning is like
engagement,
how to foster
student,
like,
authentic academic
engagement,
like behavioral,
cognitive,
affective,
and that's
connecting motivation
and active
learning.
So,
maybe for these
literature,
maybe some
ideas from

those free energy
principles,
maybe somehow
be applicable.
Also,
third,
the free energy
principle will
provide valuable
insights and
methodologies that
could be applied
to elaborating
the existing
findings and
hypothesis
hypothesis in
foreign language
studies.
Like,
just my
example,
the emotion
ball processing
hypothesis,
actually,
in a more
comprehensive
and systematic
manner.
And maybe
I'm talking a bit
about that first.
Again,
slightly off topic,
but just one
theory,
because that's

related to
my own
study.
using some
depth of
processing
model in
psychology.
I,
several years
ago,
proposed those
emotion ball
processing
hypothesis,
which is
basically about
the processing
is deeper.
When you're
just looking at
thinking about
those perceptual
aspects of
the target
stimuli or
item,
maybe the
learning is
not very
deep,
leading to
weaker memory.
But if you
think about
the meaning,
that's stronger.
That's some

established
psychological
finding,
though.
But if you
make connection
with those
emotional
aspects,
for example,
like learning
the word
like apple
in a foreign
language,
maybe if you
for example,
think about
trying to
write about
that using
the favorite
apple in
your life,
for example,
or maybe
something that's
related to
your emotional
capacity,
maybe that
learning will
be more
organically
integrated in
your cognition
and you
will be

more likely
to learn.
So this
idea is out
there,
but what
is depth?
Maybe it
needs to be
elaborated more,
maybe free
energy principle
can be of
health there
as well.
And there's
another theory
that's based
on the first
one is like
deep positive
hypothesis.
Well,
simply,
positive emotion
is important.
Higher cognition,
academic learning,
enjoyment,
or positive emotion
is more important
than negative emotion.
That's the idea
behind this.
That actually
kind of
is like
off-top,

related sideway,
though.
So I'm just
talking very briefly,
though.
Many previous
studies talk about
positive emotion
is important.
Negative emotion
is to be avoided,
but it's not
always the case.
Actually,
just trying to
pursue the
positive
enjoyability,
enjoyment
is not always
useful.
And there can
even be some
danger there,
as published
in some
recent
journal article,
like,
Beyond Positive
Psychology,
is the title
actually there.
positive
or
negative.
So maybe
I don't

talk too much
here,
but just
positive
versus
negative,
just some
dyadic
idea of
thinking about
emotion
is not
really fruitful,
but for
higher academic
cognition,
more triadic
type,
epistemic
emotion,
intellectual
activeness,
creativity,
even
ultimately
leading to
morality
or
around
the more
education
needs to
be pursued,
and that's
going beyond
positive or
negative
actually.

And for
that,
maybe some
importance
is more
about those
epistemic
emotions
there.
Since
temporarily,
fourth point,
the deep,
deliberate,
higher cognitive
processes are
related more
closely with
epistemic
emotions than
basic emotions,
epistemic
emotions may
be more
elaborate,
deeper,
than basic
emotions in
a dynamic,
multi-leveled
cognitive
emotional
continuum
there.
And that
led to
the
proposition

of
this
deep
epistemic
emotion
hypothesis.
That's
actually
my
paper.
Intellectual
learning and
acquisition
would
benefit
greatly
more by
mindfully
harnessing
those
epistemic
emotions
like surprise,
curiosity,
or even
some
temporary
confusion,
or maybe
utilizing
the
boredom.
A lot
of people
say
boredom
is not
good,

so boredom
is a
negative
emotion,
which is
kind of
true,
but even
boredom
could be
used
to
shift from
exploitation to
exploration
using
curiosity.
So not
just
enjoying
activity,
but more
about some
activity that's
making you
surprised,
making you
curious.
These will
lead to even
better learning.
theory,
so that's
theory based
on those
expected free
energy idea.
And there

is a couple
of some
papers there
as well,
so 2023,
like the
mental lexicon
paper,
there's some
empirical,
experimental,
psychological
paper,
that's basically
one of the
conclusions is
that some
incongruence-based
surprise may
lead to better
memory formation
of foreign
language wars,
and also
a forthcoming
paper,
I talk about
these three
different
hypotheses,
emotion,
positive emotion,
and beyond
epistemic emotions,
which will be
published soon
in a few
months,

I guess,
in second
language,
so more
on those
things,
those papers.
Okay,
maybe I should
wrap up
to the final
point.
Wrapping up
the discussion
so far,
free energy
principle is
compatible with
four e-cognition
aspects,
which is very
important.
now,
or more
overcoming
the problems
of the
previous
paradigms.
Prediction is
everywhere,
including in
input processing,
even consciously
and even
unconsciously.
Perception input
and action output

are inseparable,
and so do
listening,
reading,
speaking,
and writing.
There's a trade-off
between accuracy
and complexity,
and potentially
even fluency,
so they need
to be
carefully
theoretically
formulated,
and also the
relation should
also be
mindfully
harnessed.
Free energy
principle can
integrate the
temporal depth
and future-oriented
prediction,
expected free
energy,
and exploitation
from exploitation
to exploration,
and intrinsic
motivation,
exploration,
emotional
valence,
like positive,

negative,
epistemic emotions.
These are
interconnectedly
explainable
in a unified
manner
and a free
energy principle
framework,
which will
have great
theoretical insight
in any other
fields,
including language
education as well.
Envoir,
also to
an initial
theory.
Well,
this is a bit
more poetic
or theoretical
though.
Anyway,
instead of
constructing a
map of
cognition by
assembling,
building blocks
of separate
cognitive
functions,
like perception,
action,

planning,
emotion,
and so on,
the free
energy principle
elucidates the
covert first
principle that
is not salient,
difficult to
see,
or even
that's impossible
to see,
but something
at work,
that permeates
cognition from
which the
distinctified
cognitive
functions can
be explained
from.
Indeed,
how you
perceive and
interpret the
world depends
on how you
have learned
to conceptually
separate the
world.
Like,
for example,
these perception,
action,

planning,
emotion,
these are
famous,
like,
William James
category of
psychological
faculties,
but,
for example,
perception is
there.
Like,
this chapter is
talking about
perception,
this action
or something.
So,
a lot of
psychologists
just took
it granted
that these
are completely
separate
mental faculties
and then
just study
separately
and,
like,
talking about
how they
are related
with each
other,

but actually
maybe something
related with
each other
is at first
and how
we just
cut,
how we
just
theoretically
divide the
holistic
world,
maybe that's
more important
aspect.
Another
free energy
principles
of
initial
approach,
dominant
existing
classifications
are revisited,
actually.
Like,
for example,
perception
action
because
seemingly
distinct
functions
may,
in fact,

be just
different
sides
of the
same
coin
feeling
the
same
underlying
imperative
and
mechanism.
And another
thing can be
also said,
something that
looks pretty
similar,
but actually
the background
mechanism
is actually
distant.
That's also
possible.
The abstract
first principle
that permeates
seemingly
separate
entities
bears a
striking resemblance
to the
multidimensional
implicate
order

and its
elusive
hollow
movement
dynamic
wholeness.
I'm using
the word
from David
Baumian
quantum
mechanics,
but also
I'm recently
more and more
aware that
this is also
pretty similar
to those
extended
Henri

Berksonian
approaches
as well.
So,
there are
a lot
of hidden
gems
in some
classic
physics
or even
classic
philosophy
there as
well.

Recently,
I'm more
into Henri
Berkson,
so maybe
I should
have
included
here,
but at
least
for the
paper
I talked
about
Bohm.
Hollow
is like
hollow
movement,
so they
organically
interconnected,
intertwined
with each
other,
those
older,
I would
say.
Likewise,
the
conventional
basic
factors in
foreign language
education
and learning

may in
fact be
arbitrarily
separated
aspects
in an
inertial
interpretive
system,
such as,
for example,
the fundamental
four
skills.
skills.
So,
again,
as I
talked about
that before,
already,
reading,
listening,
writing,
speaking,
four skills,
and then
how they
are related
with each
other,
for example,
the score
of speaking
tests,
score of
reading
tests,

and then
how they're
correlating
with each
other,
or how
which one
is predicting
the other
or something.
Of course,
originally,
maybe the
theorists
didn't talk
about that,
think about
that,
but usually
it's
easy to
just think
about
we have
different
those
skills,
and then
we just
mix them,
or we
just pay
attention to
these distinct
ones separately,
and then
we just
try to

integrate it.
But
estimating
the
professions,
yeah,
so we
tend to
think about
that,
for example,
language
proficiency
is a
mixture
of
vocabulary,
grammar,
knowledge,
all these
four skills.
But is
this view
really
true?
Maybe
not.
And
maybe
not.
In fact,
actually,
even those
traditional
listening,
reading,
speaking,
writing,

skill
view
was
criticized
and even
revisited
in
Europe,
for example,
like Cephal,
CV,
European Common
Framework,
and in that
they
newly
recategorize
it into
reception,
production,
interaction,
which is
paying more
attention to
this
permeation of
input and
output,
and
mediation
as the
basic
element,
which is
a pretty
great
step forward.
But yet,

the increased
complexity with
further
subdivisions,
seven skills
or more,
with debatable
issues out there,
mycelaneous elements
in the category
of mediation.
and
here's
a list
of
this
Council of
European
2020
Cephal
CV
companion
volume,
a new
edition
there.

reception is
oral comprehension,
which is
listening,
reading comprehension,
which is
reading,
production,
oral production,
which is
speaking,

written production,
reading,
and interaction,
which is good,
like oral interaction,
speaking,
but like online
as a different thing.
And so mediation,
mediation is very important,
but it's kind of like
mycelaneous,
and there are a lot of
some categorical
overlapping there,
unfortunately.
And of course,
these things are
pretty useful
as some
lubricant
or to
think about
different aspects
for assessment
and stuff.
But actually,
as the free energy
principle suggests,
simple and accurate
hermeneutic
frameworks
and guidelines
will have
better pragmatic
values there.
Just
subdividing it

into many
different things
and trying to
conceptualize
like proficiency
or something
using those
as a mixture
of different things
may not
always be
a pretty sound
ontology
or epistemology.
Indeed,
supposing a fixed
set of basic
categories
and classifying
elements
like sorting
files
inside folders
approach
cannot help
being arbitrarily
to some extent.
And it's
changing.
The new theories
out there,
the old category
is no longer
making very good sense,
so you need to
constantly update
those categories
and the old theory

may be pretty
obsolete,
actually.
In reality,
however,
these elements
are organically
interconnected
with each other
in varied
qualitative
and quantitative
degrees,
or more precisely,
these categories
and elements
are conceptually
carved out
of the undivided
whole and moving
wholeness
of the implicate
order.
So maybe this
is more true.
Something is there
which is very
difficult to approach
and then
how we
epistemologically
separate things.
So that's
what is really
happening.
It should be.
Free energy
principles

ab initio
approach
having some
resemblance.
it can work
as a potentially
more fruitful
alternative
to the
sorting file
inside
folders
approach.
And also
almost the end.
Sorry,
I mean,
talking too much.
And an ab initio
approach will
start by
trying to identify
the underlying
overt mechanism
that permeates
everything relevant
in the target
of the model
via inductive
as well as
abductive reasoning
that is compensated
by epistemic
emotions there.
Surprise,
curiosity,
these are
related to

some abductive
reasoning.
Charles Perce,
the father of
American pragmatics
and many other
things,
like polymath,
one of my
favorite philosophers
in addition to
Bertheson and
Whitehead.
And also
in turn,
different functions
will be deductively
situated in the
unifying model
in an organic
manner,
which could then
be tested
and validated
empirically in
the community
of inquiry,
of scientists
and even
educators.
The presupposition
of a first
principle also
echoes those
Percian account
of truth.
Now,
a lot of people

say like the
post-truth world
or there's a lot
of different
truth,
every truth is
true,
but Berthes'
idea is like
there's some
ultimate truth
unattainable,
but we can
approach it.
And that can
be done
scientifically
through the
cognitive inquiry.
And those
truths may be
something which
is related to
those not
some covert
first principle
or of
initial approach
theory.
Because keeping
those Percian
fallibilistic and
synechistic
spirits,
which is like
we will,
our understanding
of something

which is wrong,
it must be
wrong,
so it must be
updated in the
future.
And synechistic,
everything is
actually connected
with each other,
so not to
miss or not to
forget about
this interconnectedness
of things,
which we need to
simplify when we
theorize stuff.
Those in mind,
starting an inquiry
in an
of initial
approach will
provide a
worthwhile
alternative
research program,
which will
eventually lead
to the discovery
of the
holistic,
unifying
theory that
has yet to
come in
foreign language
education field,

which will
unfold the
underlying
implicate
order in
a creative
manner that
maximizes both
epistemic and
pragmatic values
like exploration,
exploitation,
and it will
be accomplished
without losing
spirit of its
ontologically
enfolded
dynamic wholeness
and a
permuting
follow-up
movement.
I am being
more
volum here
though.
So just
some schematic
image here.
So behind
those reading,
listening,
mediation,
writing,
speaking,
everything,
there's some

unifying principle
out there,
which is very
difficult to
approach,
maybe theoretically
only you can
approach,
whether that's
really real,
the realism of
the approach,
maybe it could
be debatable,
but still,
starting with
trying to have a
look at something
hidden,
some unifying
principle,
and trying to
figure out what
that is,
and trying to
update the
theory,
that's also
another pretty
useful,
helpful approach
or the
research program
for the future.
So that's
basically it
though.
Well,

that's basically
the end.
That was
my conclusion
of the paper,
but adding
a little bit
of some new
things,
especially
the audience
for this video,
maybe now
or later,
may include
some language
educators there
as well,
maybe who have
no knowledge
about free
energy.
So for that,
maybe just one
simple key take
home could be
just setting
optimally
challenging tasks
that evoke
those epistemic
emotions like
surprise,
curiosity,
to foster
some exploration
and also
intrinsic motivation.

And that
will like to
appropriately
induce
prediction errors
in learners
and generating
interesting
curiosity
through
and that
can be done
by some
interactive
inquiry-based
dialogic
approaches
and models
there.
And here's
just my
several examples
that you
call
links are here.
So like
challenges
reading circle,
like this is
some group
based
reading,
active learning
approach that
you ask
questions with
each other
in a structured

manner and
then usually
that leads
to some
surprise,
curiosity
or like
tip of the
tongue
effect.
And this
usually lead
to better
memorability
and learning.
P4ELT
based on
P4C
philosophy
for children
college
community
approach,
which is
based more
on the
authentic
dialogue,
something that
you're really
interested in,
you discuss
with each
other
democratically.
So this
idea,
applying it

to English
language
teaching
is a
pretty
useful,
maybe
one way
to
operationalize
it.
Case
teaching,
ELT,
case teaching
originally
some business
school
thing,
but using
some real
scenario
or like
pseudo
scenario
and then
you discuss
as if
you were
in the
situation
and to
invigorate
intrinsic
curiosity
and
authenticity
and just

it's a bit
some 10
minutes short
activity stuff
but IPA
news
just talking
about some
news article
stories
in some
unique
pronunciation
symbol
that's
authentic
so usually
instead of
just learning
the symbol
as it is
just
situating it
in the
real
life
events
and then
making it
as a
quiz
is more
interesting
motivation
inducing
and
curiosity
inducing

theoretically
too
it's quite
recent
though
I talked
about
briefly
complex
dynamic
system
theories
language
learning
which is
highly
studied
this idea
about
those
trying to
look into
those
dynamic
interplay
of different
factors
that's
another
very
promising
approach
in
foreign
language
education
study
and

this year
also
last year
proactive
language
learning
theory
the new
model
in second
language
acquisition
according to
that
input
seeking
behavior
so it's
not just
listening
or reading
but even
behind listening
and reading
there is
this
agentic
approaches
for each
learner
so we
actually
seeking
for those
input
and
so this
is based

on the
idea
that
everything
is actually
kind of
like a
top-down
manner
so there
are more
and more
models
that's also
trying to
capture
the spirit
of this
some
insight
echoing
the insights
of free energy
principles
and again
for the
reading
so most
of the
talk
today
is based
on this
so here
is the
link
and
URLs

there
and maybe
I'm updating
the research
maps
things
so maybe
there may
be some
new
information
there
from time
to time
a little
bit of
promotion
there
I'm
also a
coordinator
leading
coordinator
of this
language
education
technology
fundamental
theory
special
interest
group
which is
a
Japanese
Kansai
based
small

research
group
that's
mostly
we're
talking
about
some
foreign
language
education
related
technology
psychology
theory
practices
so
basically
every
two
months
we
are
having
zoom
or
in-person
meetings
there
so
some
x-link
or
website
there
and also
we are
having

another
meeting
in
Tokyo
at the
end
of
this
month
and
stuff
also
that's
also
on
zoom
so
if you're
interested
don't
hesitate
to just
follow
these
for
more
okay
I think
this is
basically
what I
prepared
for
today
as I
have
more
information

also
like
KNZW783
on
X
or
blue
sky
you
will
find
several
additional
resources
and
other
things
too
so
hope
I
kind of
like
try to
integrate
or
incorporate
make
connection
with
different
ideas
and
maybe
the
study
application
for

the
future
is
up
to
the
future
study
and
research
and
others
so
also
good
to
keep
discussion
or
interdisciplinary
thinking
and even
potential
collaboration
stuff
though
okay
that's
about
it
thank
you
so
much
awesome
thank
you
okay

for those
watching
live
they can
write
any
questions
I
kind
of
wanted
to
pick
up
on
the
connection
with
David
Bohm
there
you
brought
up
communication
and
the
underlying
sort
of
pilot
wave
of
proficiency
in terms
of the
implicate
order

but
Bohm
also
developed
the
dialogue
style
so
it's
interesting
how did
you come
to that
connection
and what
do you
think
that
means
that
there's
sort
of
like
a
physics
of
cognition
that's
reached
through
dialogue
and
also
vice
versa
yeah
yeah

so
basically
the idea
that I
the
implicate
order
that's
the word
that I
thought
is the
best
fit
to
explain
what I
wanted
to
talk
about
but
of
course
like
late
David
Bohm
talked
about
that
importance
of
dialogue
and
others
too
which

is
actually
pretty
much
echoing
those
Charles
Percy's
community
of
inquiry
and
also
what was
that
P4C
that I
talked
about
P4ELT
is
another
way
I
put
it
though
so
through
dialogue
discussion
and
then
together
you
approach
closer
to

that
truth
or
some
true
cosmos
I would
say
something
hidden
kind of
implicate
order
which is
dynamic
so
if we
just
say
this
is
it
so
we
verbalize
everything
and this
is
the
whole
rule
maybe
this
is
not
always
true
which is

actually
impossible
to
just
reach
so
you
need
to
be
fallibilistic
though
so
in that
sense
I didn't
actually
I read
but
I haven't
read for
long
about
those
dialogue
aspects
but
that's
pretty
much
resonating
with
those
Persean
idea
and also
this
committee

of inquiry
philosophy
for
college
or
English
language
education
idea
there
so
yeah
that's
kind
of
mutual
permeation
there
theoretically
as well
so
the
direction
is
basically
similar
yeah
that was
awesome
you made
many
great
connections
there
so
how
do you
think

first
and
second
language
learning
differ
and
how do
you
approach
that
or
how
is
it
studied
how
does
FEP
help
us
look
at
that
because
it seems
like that
first one
comes pretty
easily
but
second
languages
can be
quite a
struggle
and there's
connections

with
developmental
critical
periods
and all
of that
but
how can
we look
at that
yeah
thank you
so yeah
a lot
of
studies
debate
disagreeing
point
about those
things out
there in
my field
for example
there's a
critical
period
some people
say
we have
critical
peers
so if
you're
older
and
this
age

maybe
you
can
no
longer
acquire
those
native
fluency
but
some
people
also
say
that
well
this
is
just
sensitive
period
only
so
there's
no
clear
critical
period
out there
and also
there's
even
now
especially
in applied
linguistics
there are
more

some
social
justice
terms
some
people
even
say
first
language
second
language
division
we
need
to
stop
so
of course
!
native
speakers
you
speak
native
even
those
can
lead
to
discrimination
or
overlooking
that
diversity
but
going there
too much

will
make it
less
scientific
actually
especially
if you
go too
far
into
there's
no
first
language
second
language
but
still
cognitively
speaking
there's
differences
out there
like
bilingual
for
example
if
you
are
born
with
two
or
three
languages
at
the

same
time
maybe
the
second
language
first
language
are
kind
of
mutually
affecting
each
other
but
if
you
have
your
own
first
language
only
one
and
then
you
just
started
to
learn
at
school
for
example
at
the

age
of
12
or
something
maybe
that
needs
to
be
different
yeah
mechanism
out
there
so
in
that
sense

I
completely
making
all
languages
equal
or
just
not
disregarding
the
first
language
second
language
but
still
especially

for
that
this
idea
of
free
energy
principles
based
insights
will be
useful
so
first
language
acquisition
if
it's
the
first
language
like
you
are
like
baby
or
like
first
year
old
second
year
old
maybe
you
don't
you

don't
think
you
just
intuitively
or
like
maybe
without
even
aware
of
those
like
higher
cognition
whatever
you
just
acquire
like
input
flood
and
also
you
kind
of
is
that
echoing
or
that's
kind
of
you
simultaneously
learn

the
world
as well
so
that's
more
organic
process
so
which
will
be
part
of
you
so
maybe
it's
effortless
process
though
but
the
second
language
especially
if
you
learned
it
at
school
not
in
communication
stuff
there
so

you
need
different
approaches
there
especially
so
for that
maybe
just
one
typical
traditional
way
just
focusing
on
okay
so
let's
learn
input
first
a lot
of
input
and
then
you
now
you
just
move
on
to
output
or
like

okay
so
you
have
different
skills
set
up
skills
so
let's
learn
these
skills
separately
which

is
very
operationalizable
understandable
and
easier
to
just
conceptualize
but
need
to
be
careful
that
how
to
operationalize
it
and
how

it
really
is
should
be
different
so
in that
sense
there
can be
actually
completely
kind of
it's
very complex
and
mutually
intertwined
with each
other
even
at
some
counter
intuitive
way
sometimes
so
it's
always
good
to
just
keep
in
mind
about

those
dynamic
interplay
and
that's
as
I would
say
kind of
ontologically
speaking
or epistemologically
speaking
and also
for those
learning
activity
as well
so
kind of
like
using
the boost
of
emotion
I would
say
especially
relating
to
those
expected
free
energy
minimization
exploration
idea
there

so
those
surprise
surprise
all
is
to
be
minimized
for
the
variational
level
but
expected
level
even
those
surprise
may
lead
to
exploration
curiosity
and
more
engagement
and
that's
really
useful
maybe
that's
also
fitting
well
into
how

our
brain
works
about
the
really
authentic
drive
toward
learning
so
maybe
learning
activity
in
second
language
maybe
somehow
maybe
it can
be
artificial
but
hopefully
authentically
made it
into
infusing
some
authentic
curiosity
or
some
surprise
element
maybe
you

are
surprised
about
something
you
read
for example
some
reading
passage
right
so
you
are
internally
curious
about
that
like
there's
some
cognitive
dissonance
or
something
you
want
to
know
more
you
want
to
see
what's
happening
here
or

something
so
those
are
actually
better
than
okay
just
read
this
and
just
answer
this
question
something
so
those
curious
stories
not necessarily
happy
story
may lead to
some
deeper
thinking
without
just
exercising
will
I
need
to
study
language
or

just
you
internally
intrinsically
motivated
to learn
something
with
surprise
curiosity
so
maybe
trying
to
prepare
some
activity
tasks
materials
just
figuring
those
epistemic
emotions
I think
will be
especially
useful
as a
second
language
term
for
some
material
developer
or
a teacher

they can
think
about
yeah
sometimes
it feels
like a
flood
of
surprises
when
it's
just
it
it
it's
like
being
with
it
and
then
losing
the
grasp
but
I
guess
that's
what
the
proficiency
is
is
being
able
to
stay

with
the
flow
of
the
sights
and
the
sounds
and
until
it
becomes
incorporated
into
a
subliminal
level
right
um
what
kinds
of
empirical
studies
or works
in your
group
or
or
um
kind
of
data
sets
that
shine
a light

on
what
kinds
of
language
exercises
or
patterns
like
we can
analyze
this
way
okay
so
empirical
studies
actually
my empirical
studies
are not
very
pedagogical
studies
actually
my empirical
studies
has
mostly
I
you know
my
one of
my
research
areas
vocabulary
so

I
have
uh
usually
done
experimental
studies
about
some
vocabulary
acquisition
like
using
computers
and
then
different
conditions
of
presenting
sentences
or words
different
tasks
to like
process
put out
and also
like answer
and there are
different sides
of keyboard
like
yes
no
or
choose
the best

one
something
and then
after that
they are
tested
the
like
else
about
the memory
of those
study
session
items
and
actually
which
condition
resulted
in
better
memory
and also
for example
this one
on the
right
side
this one
is like
some
negative
sentence
with
positive
word
or

positive
word
negative
sentence
so
some
condition
that
makes
you
somewhat
surprised
like
what
so
like
some
makes
you
feel
curious
actually
these
conditions
led
to
better
memory
performance
although
it was
incidental
learning
condition
in
other
words
students

were not
instructed
to
try to
memorize
everything
so
it was
only
when
they
saw
it
just
once
or
twice
only
and
naturally
they
remembered
some
sentences
that
had
some
surprise
in
it
so
in
that
sense
this
is
another
empirical

side
of
the
deep
epistemic
emotion
hypothesis
actually
though
so
empirical
study
this
is
it
and
of
course
something
that's
coming
before
is
like
this
is
like
some
shallow
processing
deep
processing
task
so
some
condition
that
you

need
to
think
about
the
meaning
carefully
and
that
was
facilitated
more
by
some
positive
words
or
positive
context
so
that's
actually
positive
emotion
task
and
this
is
another
similar
study
one
of
the
oldest
one
like
some

similar
condition
like
apple
looking at
some
easy
english
word
english
is
a foreign
language
word
for
japanese
student
and
just
they
look at
the
word
they
think
about
the
emotion
aspect
they
think
about
the
meaning
aspect
or
the
perception

aspect
the
emotion
aspect
resulted in
deeper
so
these
are
there
and
about
pedagogy
pedagogically
speaking
maybe
several
insights
here
maybe
I need
to do
more
empirical
study
about
there
there
are
still
kind
of
like
some
practical
ideas
with
some

student
responses
I
published
some
links
information
available
here
but
usually
that
leads
to
students
enjoying
or
engaged
and
also
they
learned
they
are
more
interested
in
what
they
learned
or
they
are
kind
of
a
flow
state

sometimes
but
it's
kind
of
hard
to
experimentally
test
the
memory
of
specific
items
especially
because
they
may
encounter
the
same
item
in
different
situations
outside
classes
and
stuff
as
well
but
at
least
some
psychologically
speaking
activity

does
leading
to
natural
engagement
natural
interest
and
motivation
surely
known
to
affect
better
learning
performance
and for
that
maybe
those
epizemic
emotions
based on
those
expected
free
energy
minimization
and also
the
interconnectedness
of
different
factors
will
provide
a good
insight

about that
which
I'm
still
on it
so
I
haven't
graphed
everything
yesterday
I was
reading
those
naive
2020
this
year's
new
book
and
others
so
maybe
I
may
need
to
myself
be
fallibilistic
and
try and
update
my
way of
thinking
or

putting
it
into
hypothesis
and stuff
yeah
that
feels
really
important
I mean
if
the
examples
or
the
test
questions
in
language
learning
curriculum
can be
brought
into
something
like a
zone
of
optimal
surprise
or
there
are
some
exercises
or
prompts

that
learners
could
be
guided
through
like
rather
than
just
reading
a
vocabulary
list
like
each
term
you know
just like
you had
with the
apple
there
each
term
do
this
sequence
of
events
when
you
see
the
term
here's
kind
of

the
pattern
of
how
you
process
it
internally
that
could
bring
a lot
of
cognitive
neuroscience
to bear
on
what
otherwise
is
kind
of
this
implicit
skill
that
it's
just
like
am I
supposed
to
look
at
this
list
of
words

and
just
memorize
it
but
even
if
I
wanted
to
do
that
like
definitely
in
school
and
even
now
in
language
learning
sometimes
I
just
stare
at
words
I
just
think
what
am
I
supposed
to
do
just

look at
this
until
what
happens
just
wait
wait
for
lightning
to
strike
or
something
but
this
gets
really
into
the
procedural
and
into
even
holistic
and
integrative
ways
to
connect
it
with
like
with
our
embodiment
that
could

ground
language
learning
and
bring
something
that
can
be
really
tedious
and
cognitively
demanding
into
something
that
like
could
and
should
connect
with
what's
delightful
and
what's
motivating
with
why
we're
learning
this
language
and
also
you're
having

some
better
idea
about
cognition
so
instead
of
just
separating
scale
trying
to
improve
each
one
so
there
is
some
interconnectedness
behind
and
also
some
ideas
in
free
energy
principle
which
will
provide
sound
more
I would
say
of course

all
theories
may need
to be
revisited
somehow
but
something
which
is
more
scientifically
grounded
and
also
more
holistic
I would
say
and
also
more
taking
into
account
those
permeating
nature
of
different
things
kind
of
like
a
Bergsonian
sense
and

so
that's
we'll
provide
some
maybe
better
even
for
for example
like
language
educator
education
for example
that's
also
what
I'm
involved
into
future
language
teachers
teaching
them
in
graduate
schools
for
them
to
have
of
course
the
classic
theories

of
language
learning
is
nice
good
to
know
as
classics

but
maybe
actually
things
are
more
complexly
intertwined
with
each
other
or
actually
we
are
kind
of
unconsciously
making
predictions
all
time
or
so
those
things
it's

good
for
also
future
teachers
to
at least
know
and
try
to
see
how
they
can
operationalize
it
also
like
when
they
look
at
students
they
can
also
think
about
those
dynamic
things
going
on
or
like
they
can

assume
maybe
better
epistemology
or better
way of
looking at
human
cognition
and
that
will
result
in
better
so
why
you
can
see
how
to
motivate
students
maybe
looking
at
into
these
for example
epistemic
emotions
and other
stuff
will
provide
better
insight

or
better
approach
about
how
to
make
students
motivated
and
stuff
instead
of
just
okay
motivation
factor
here
learning
factor
here
how
to
combine
but
maybe
something
behind
is
actually
pretty
much
connected
so
maybe
working
on
that

looking
at
that
is
really
important
for
educators
as
well
also
I
think
you
very
richly
entered
into
these
philosophical
questions
and
I
also
thought
about
the
generative
modeling
of the
student
and
also
the
language
learning
of
language

models
and other
kinds of
artificial
intelligence
and how
they have
some
similar
dynamics
and
yet
you
know
did
they
also
have
free
energy
principle
grounded
language
learning
or what
is
human
language
learning
is
what
we
engage
in
but
what
might
we

learn
from
other
systems
and
their
language
learning
and
like
would
their
would
the
generative
models
that
describe
their
proficiency
have a
similar
templating
or
perhaps
different
templating
than
ours

well
so
maybe
I
think
maybe
not
completely

digesting
it
myself
but
maybe
some
Markov
blanket
with
some
autonomy
element
is
needed
or
energy
information
structure
adaptation
closure
or
constraint
closure
I
think
there's
something
there
that's
maybe
better
that
we
can
incorporate
more
from
it

actually
so
some
autonomousness
intentionality
so
these
needs
to be
more
theoretically
incorporated
so
in that
sense
maybe
this
is
not
negating
free
energy
but
expanding
or
maybe
sophisticating
free
energy
insights
in
bio
inactivistic
ideas
out there
so
I
think

especially
for
education
those
bio
inactivistic
ideas
will
probably
be
more
useful
so
in that
sense
maybe
there's
something
more
I
need
to
learn
as
well
so
I
cannot
make
a
pretty
short
comments
but
I
think
this
free

energy
principle
is
onto
something
a
very
good
something
I
would
say
and
also
that's
surprisingly
echoing
some
old
insight
like
I
might
say
like
some
like
Berkson
whitehead
purse
I
think
you know
like
maybe
people
will
know

like
oh
you
like
those
philosophers
type
of
vision
I
have
but
surprisingly
those
free
energy
principle
having
some
shared
view
like
the
interconnectedness
or like
a
perception
action
extension
or
trying to avoid
just fragmenting
things
trying to
grasp
ungraspable
wholeness
and then

how to
derive
different
mental faculties
or functions
out of it
in a very
dynamic
manner
manner
and
just
even
languages
itself
can
kind
of
fix
it
and
stabilize
concepts
and
they may
become
easily
become
some
dead
concepts
and
if
you
just
use
it
apply

differently
maybe
it's
no
longer
very
valid
those
validation
crisis
is a
pretty
big
thing
in
my
field
some
for
example
motivation
theory
a lot
of
people
talk
about
that
use
that
theory
but
actually
maybe
that's
not
very
scientifically

valid
so
those
claims
are
made
but
especially
for
those
theories
models
to
propose
that
based
not only
on
observations
or
some
factor
analysis
and
studies
but
something
based
more
on
those
philosophy
I would
say
or
even
some
deep

neuroscientific
including
free
energy
but
also
even
some
Bergsonian
Percian
Whiteheadian
ideas
about
so
like
human
nature
even
some
baby
momian
idea
of course
in Canada
as well
will
maybe
provide
some
more
like
long
lasting
research
framework
and
that'll
be

pretty
more
useful
than
just
something
here's
a new
category
and
stuff
so
yeah
so
that's
the thing
it's a
great
point
and
language
itself
and
the
study
of
language
revealing
our
ontologies
for
the
world
and
these
philosophical
topics
of

you know
nouns
and verbs
and adjectives
and grammar
and
like
using
language
to learn
about
language
including
what
the
reference
of
language
is
and
like
that
sort
of
inactivism
meets
semiosis
and
the
self
awareness
of that
process
and the
science
of that
process
yeah

semiosis
exactly
that's
first
yeah
like
tri
is not
just
two
what is
that
caesurian
like
sinipion
siniphea
but like
he's like
interpreting
the third
kinds
out there
which is
kind of
similar
to
dialectics
there
but
that's
pretty
also
onto
something
and also
at the
same
time

it's
kind
of
like
a
new
project
that
I'm
kind
of
starting
to
work
on
but
of course
looking
into
those
universal
like
rule
behind
something
is also
important
but also
at the
same
time
especially
language
there
are a lot
of
some
diversities

for example
I'm based
in Japan
so like
some
Japanese
language
or some
Japanese
psychological
background
and also
for example
like
Germanic
for example
or
British
or even
like
Europe
there are
a lot
of
differences
out
there
Africa
different
parts
of
Asia
America
Latin
North
so
differences
out

there
but
in
those
differences
how
things
are
different
but
still
how
they
are
similar
so
those
are
also
pretty
much
interesting
to look
into
even
like
language
teaching
approaches
like
that's
pretty
those
cultural
variations
are
pretty
big

like
Japanese
shy
students
tend
to be
shy
and
perfectionist
or
!
differences
surely
will
make
some
difference
out
there
but
still
some
cultural
concepts
could
also
be
applied
from
one
culture
to
different
something
that
I'm
just
publishing

a paper
soon
is
something
called
Yari-gai
which
is
based
on
Ikigai
is
kind
of
buzzword
worth
worthness
of
living
or
something
but
that's
situating
into
activity
is
something
called
Yari-gai
and
that's
actually
a lot
of
several
studies
are

published
already
in
English
about
what
constitutes

Yari-gai
which
is
Japanese
concept
but
that's
kind
of
integrating
engagement
motivation
also
not
just
that
some
value
or
social
contribution
a lot
of
different
things
inside
there
so
something
I'm

also
trying
to
do
is
how
to
grasp
those
interconnectedness
using
some
nice
keywords
so
something
that
I'm
starting
is
using
Yari-gai
that's
a
paper
I think
soon
to be
published
in May
or
something
but
yeah
so
that's
another
way

going
to
some
are
related
not
completely
that
sounds
very
interesting
thank
you
yeah
well
do you
have
any
last
comments
or
updates
or
what
directions
you're
going to
be
taking
things
yeah
so
basically
thank
you
so
I
think

I'm
still
learning
a lot
and
yeah
it's
metaphorically
speaking
it's
always
important
not
to
like
sit
deep
on
a
couch
or
something
I would
say
so
you know
I was
like
okay
so here
is
the
theory
I
have
so
I
just

live on
it
or
something
we
shouldn't
do
that
so
like
always
looking at
some
new
ways
of
the
latest
psychological
and
neuroscientific
ideas
actually
deep
philosophical
ideas
that's
kind of
forgotten
but
pretty
useful
kind of
like
driven
by
intuition
actually

in a
sense
though
but
kind of
keeping
on
learning
and also
keeping
on
even
collaboration
as well
so
actually
so
my
field
is
second
language
education
basically
but
so
a lot
of
including
free
energy
I
draw
insights
outside
second
language
education

typical
literature
actually
but
that's also
very important
there
I think
and also
even
potentially
important
for
like
from
science
there
too
for
science
to
for
example
like
language
education
is a
field
of
really
big
applicability
it's like
theory
and practice
are kind
of
mixing

with
each
other
sense
so
actually
in there
those
dynamisms
maybe
it can
be used
as a
nice
field
to
test
or
maybe
observe
some
theoretical
aspects
there
too
so
maybe
trying
to
search
for
or
maybe
even
deeper
maybe
even
creative

interdisciplinary
interdisciplinary
collaboration
or
borrowing
insight
from
the
ones
the
other
so
keeping
on
thinking
about
that
and
also
revising
it
and
trying
to
approach
that
truth
in a
Persian
sense
maybe
that's
a
pretty
important
thing
to
go

for
and
also
I'd
like
to
keep
on
the
journey
so
that's
why
also
I
started
learning
free
energy
little
by
little
now
yeah
so
I'd
like
to
continue
the
journey
and
also
I
hope
that
listeners
or

anybody
is
also
trying
to
how
it
could
be
applied
for
example
education
or
how
their
observations
can be
linked
to
some
philosophical
insights
or
even
these
neuroscientific
insights
out
there
so
trying
to
go
interdisciplinary
and
go
even

creative
will
be
something
which
I
would
like
to
maybe
as
a
final
comment
and
also
some
final
agenda
manifest
myself
as
well
as
a
researcher
I
would
like
to
just
keep
on
doing
it
awesome
I
think

where
what
you've
presented
takes
me
into
sleep
and
beyond
is
that
when
we
have
dialogue
and
went
across
our
agent
boundary
with
another
and
when
we
have
dialogue
internally
with
multilingualism
internally
there's
the
juxtaposition
of
multiple

syntaxes
that allow
for that
triangulation
of
semantics
and that
can
can
open up
a
triangle
or
more
whereas
within
one
perspective
and one
language
it's like
looking only
through
a microscope
or a
telescope
but
in
external
and
internal
dialogue
and
learning
we
can
really
get

somewhere
exciting
exactly
even
in your
brain
as well
there's
some
I'm
kind of
like a
frequent
reader
of
things
and
if
anybody
here
follow
my
X
or
something
I
usually
post
a lot
about
some
papers
that I
read
but
there
are
some

multilingual
theories
suggesting
that
even
your
personality
ways
of
thinking
is
influenced
by
the
language
that
you're
using
so
if
you
just
use
one
language
to
think
maybe
of
course
that's
fine
but
if
you
have
several
two

or
three
languages
maybe
you
have
different
insight
even
within
you
for
example
you
think
about
it
in
Japanese
language
but
the
other
day
you
think
in
English
language
different
syntax
maybe
no
complete
theory
there
I
think

that
leads
to
some
new
ways
of
thinking
even
when
you're
just
thinking
yourself
and
maybe
that's
also
good
for
brains
people
with
bilinguals
trilinguals
even
having less
dementia
risk
or
something
so
even
with
yourself
internal
dialogue
there

too
maybe
having
a lot
of
those
different
languages
second
language
foreign
language
and
try
to
use
it
in
your
brain
looking
at
the
same
thing
in
different
perspectives
or
different
concepts
overlapping
in
one
language
with
different
others

and
things
so
these
are
also
pretty
good
way
to
exercise
your
cognition
and
try
to
make
it
dynamic
that's
that
kind
of
key
word
there
so
complex
and
dynamic
so
try
to
capture
those
keep
up
not

to
fossilize
your
cognition
maybe
it's
good
to
use
maybe
shake
yourself
not
not
to
fossilize
your
brain
or
thinking
yourself
maybe
learn
new
languages
and
then
new
concepts
new
ways
of
thinking
and
keeping
on
dialogue
externally

also
maybe
internally
would be
a pretty
good
powerful
tool
those
foreign
languages
are
that's
a
thing
I
also
experientially
thank
myself
too
thank
you
again
looking
forward
to
hearing
more
updates
from
your
research
and
learning
with
you
yeah

thank

you

thank

you

so

much

bye

thank

Thank you.